

- 10 An object X has weight W on the surface of the Earth. How much energy is needed to raise X to height of $2R$ above the surface of the Earth, where R is the radius of the Earth?

A $\frac{1}{3}WR$

B $\frac{2}{3}WR$

C $\frac{3}{2}WR$

D $2WR$

Ans: B

$$W = \frac{GMm}{R^2}$$

$$\text{Increase in GPE} = -\frac{GMm}{3R} - \left(-\frac{GMm}{R}\right) = \frac{2GMm}{3R} = \frac{2}{3}WR$$

11 The density of air is 1.20 kg m^{-3} at a temperature of 20°C and standard atmospheric pressure